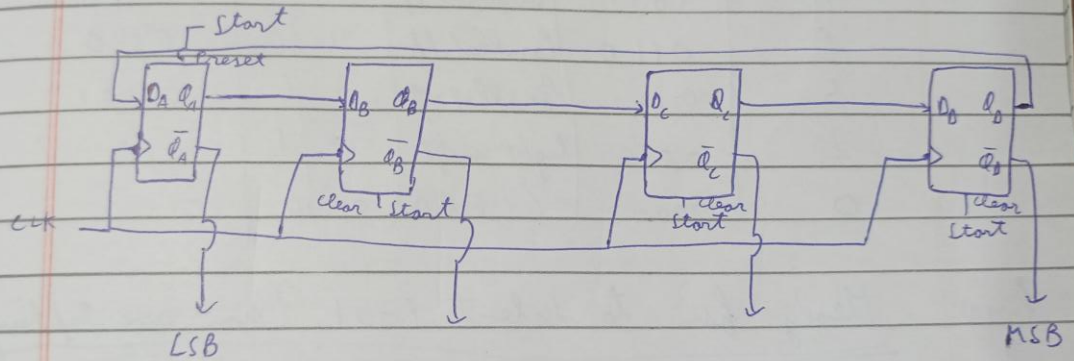


## Inside the SoC - Assignment Week-1

Q.2 Same circuit but I took  $\bar{Q}$  as output.



Here last flipflop is MSB, & every clock cycle, 0 is left shifted in our number.

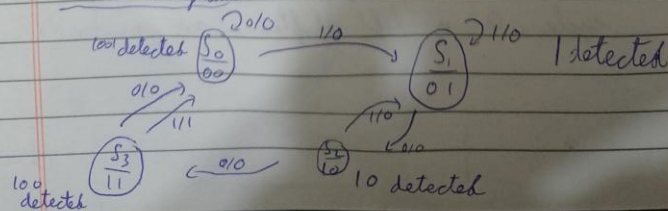
Q.3 Using USR where selector variables of four  
4x1 MUX are  $S_1$  &  $S_0$ .

Pattern to get = 0-1-2-4-9-11-6-5-2-0...

| Decimal no. | Binary rep. | Operation    | $S_1$ | $S_0$ | Parallel/serial input |
|-------------|-------------|--------------|-------|-------|-----------------------|
| 0           | 0000        | —            | 0     | 0     | —                     |
| 1           | 0001        | Left shift   | 1     | 0     | 1                     |
| 2           | 0010        | Left shift   | 1     | 0     | 0                     |
| 4           | 0100        | Left shift   | 1     | 0     | 0                     |
| 9           | 1001        | Left shift   | 1     | 0     | 1                     |
| 11          | 1011        | Parallel inh | 1     | 1     | 1011                  |
| 6           | 0110        | Parallel inh | 1     | 1     | 0110                  |
| 5           | 0101        | Parallel inh | 1     | 1     | 0101                  |
| 2           | 0010        | Right shift  | 0     | 1     | 0                     |
| 0           | 0000        | CLEAR        | 0     | 0     | —                     |

Q.4a) Mealy fsm to detect 1001 (non-overlapping) -

State diagram -



## State Table -

| Prev. states |       | Input<br>X | Next states |         | Output<br>Y |
|--------------|-------|------------|-------------|---------|-------------|
| $q_A$        | $q_B$ |            | $q_A^+$     | $q_B^+$ |             |
| 0            | 0     | 0          | 0           | 0       | 0           |
| 0            | 0     | 1          | 0           | 1       | 0           |
| 0            | 1     | 0          | 1           | 0       | 0           |
| 0            | 1     | 1          | 0           | 1       | 0           |
| 1            | 0     | 0          | 1           | 1       | 0           |
| 1            | 0     | 1          | 0           | 1       | 0           |
| 1            | 1     | 0          | 0           | 0       | 0           |
| 1            | 1     | 1          | 0           | 0       | 1           |

## Y K-map -

| $q_A q_B \backslash X$ | 00 | 01 | 11 | 10 |
|------------------------|----|----|----|----|
| 0                      | 0  | 0  | 0  | 0  |
| 1                      | 0  | 0  | 1  | 0  |

$$\Rightarrow Y = q_A q_B X$$

## $q_A^+$ K-map -

| $q_A q_B \backslash X$ | 00 | 01 | 11 | 10 |
|------------------------|----|----|----|----|
| 0                      | 0  | 1  | 0  | 1  |
| 1                      | 0  | 0  | 0  | 0  |

$$\Rightarrow q_A^+ = \bar{q}_A q_B \bar{X} + q_A q_B \bar{X}$$

$$= \bar{X} (q_A q_B)$$

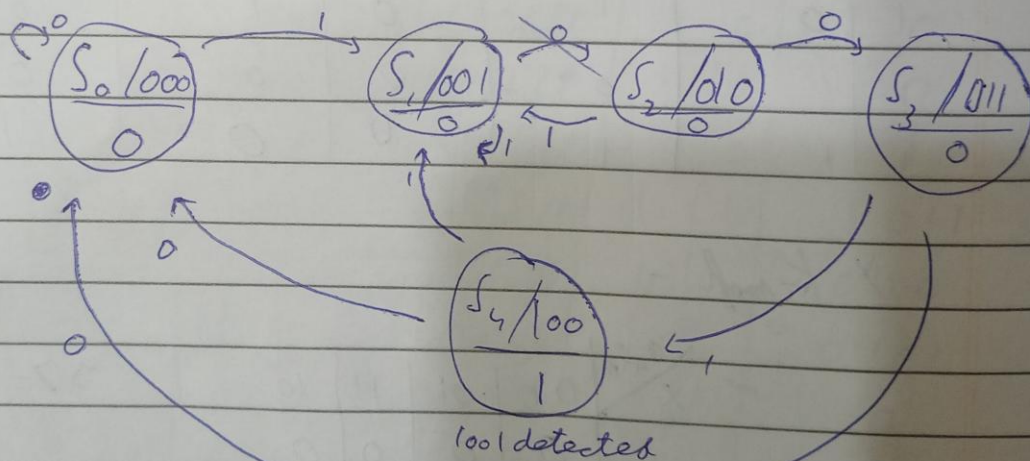
$\phi_B^+$  K-map -

| $\phi_A \phi_B$<br>x | 00 | 01 | 11 | 10 |
|----------------------|----|----|----|----|
| 0                    | 0  | 0  | 0  | 1  |
| 1                    | 1  | 1  | 0  | 1  |

$$\Rightarrow \phi_B^+ = \bar{\phi}_A X + \phi_A \bar{\phi}_B$$

Moore fsm to detect 1001 (non-overlapping) -

State diagram -



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| Prev. state |       |       | Input | Next state |         |         | output |
|-------------|-------|-------|-------|------------|---------|---------|--------|
| $Q_A$       | $Q_B$ | $Q_C$ | $X$   | $Q_A^+$    | $Q_B^+$ | $Q_C^+$ | $Y$    |
| 0           | 0     | 0     | 0     | 0          | 0       | 0       | 0      |
| 0           | 0     | 0     | 1     | 0          | 0       | 1       | 0      |
| 0           | 0     | 1     | 0     | 0          | 1       | 0       | 0      |
| 0           | 0     | 1     | 1     | 0          | 0       | 1       | 0      |
| 0           | 1     | 0     | 0     | 0          | 1       | 1       | 0      |
| 0           | 1     | 0     | 1     | 0          | 0       | 1       | 0      |
| 0           | 1     | 1     | 0     | 0          | 0       | 0       | 0      |
| 0           | 1     | 1     | 1     | 1          | 0       | 0       | 1      |
| 1           | 0     | 0     | 0     | 0          | 0       | 0       | 0      |
| 1           | 0     | 0     | 1     | 0          | 0       | 1       | 0      |
| 1           | 0     | 1     | 0     | X          | X       | X       | X      |
| 1           | 0     | 1     | 1     | X          | X       | X       | X      |
| 1           | 1     | 0     | 0     | X          | X       | X       | X      |
| 1           | 1     | 0     | 1     | X          | X       | X       | X      |
| 1           | 1     | 1     | 0     | X          | X       | X       | X      |
| 1           | 1     | 1     | 1     | X          | X       | X       | X      |

7 K-map -

| $Q_A \backslash Q_B$ | 00 | 01 | 11 | 10 |
|----------------------|----|----|----|----|
| 0                    | 0  | 0  | 0  | 0  |
| 1                    | 0  | 1  | 1  | 1  |

$$Y = Q_A \overline{Q_B} \overline{Q_C}$$

 $Q_A^+$  K-map -

| $Q_A \backslash Q_B$ | 00 | 01 | 11 | 10 |
|----------------------|----|----|----|----|
| 00                   | 0  | 0  | 0  | 0  |
| 01                   | 0  | 0  | 1  | 0  |
| 11                   | X  | X  | X  | X  |
| 10                   | 0  | 0  | X  | X  |

$$\Rightarrow Q_A^+ = Q_B Q_C$$

 $Q_B^+$  K-map -

| $Q_A \backslash Q_B$ | 00 | 01 | 11 | 10 |
|----------------------|----|----|----|----|
| 00                   | 0  | 0  | 0  | 1  |
| 01                   | 1  | 0  | 0  | 0  |
| 11                   | X  | X  | X  | X  |
| 10                   | 0  | 0  | X  | X  |

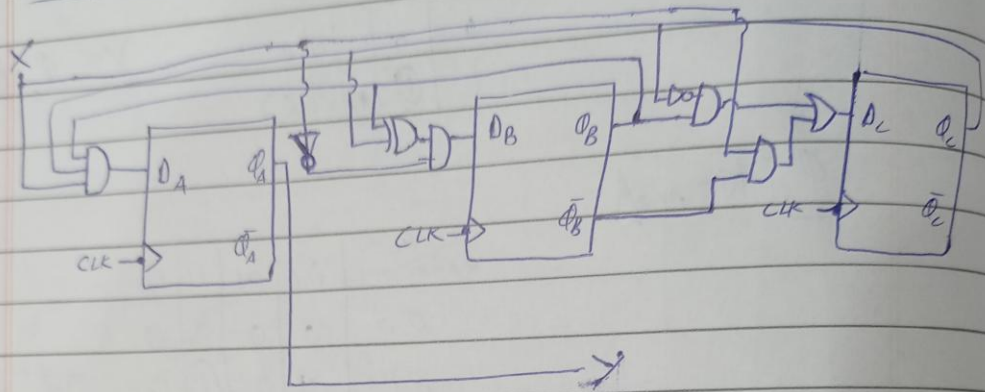
$$\Rightarrow Q_B^+ = Q_B \overline{Q_C} \overline{X} + \overline{Q_B} Q_C \overline{X} = \overline{X} (Q_B \oplus Q_C)$$

 $Q_C^+$  K-map -

| $Q_A \backslash Q_B$ | 00 | 01 | 11 | 10 |
|----------------------|----|----|----|----|
| 00                   | 0  | 1  | 0  | 0  |
| 01                   | 1  | 1  | 0  | 0  |
| 11                   | X  | X  | X  | X  |
| 10                   | 0  | 1  | X  | X  |

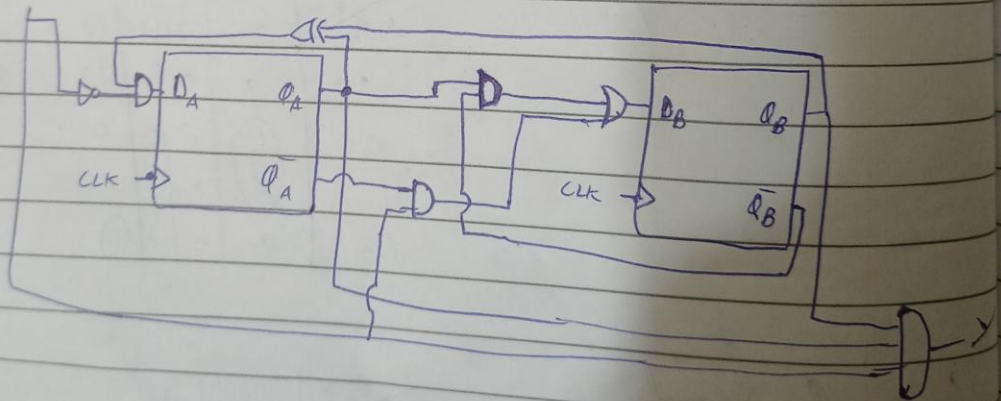
$$\Rightarrow Q_C^+ = \overline{Q_B} X + Q_B \overline{Q_C}$$

Moore circuit -

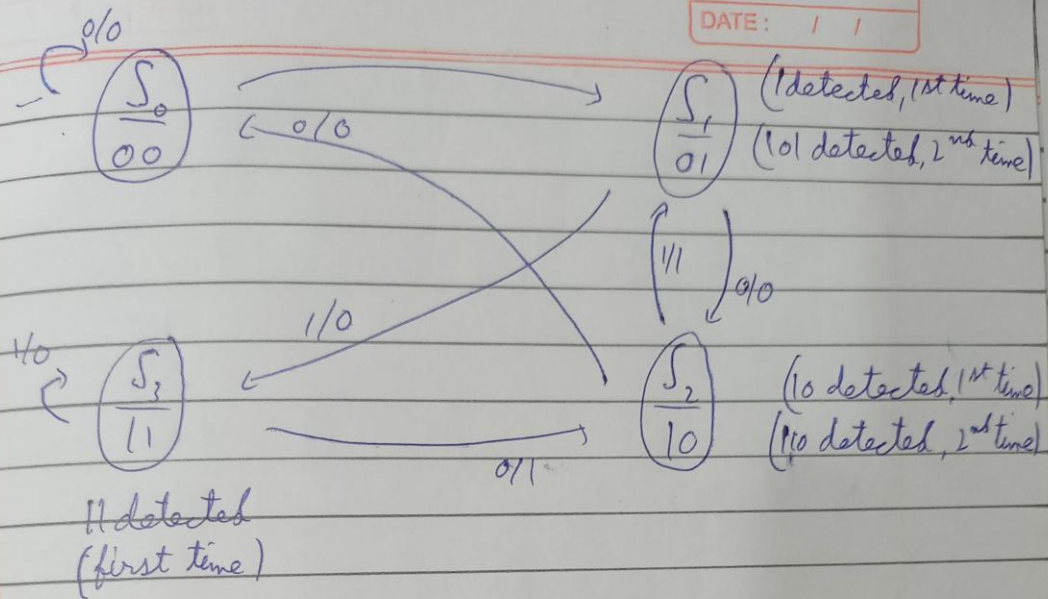


Mealy circuit -

X.



Q.4b) Mealy fsm to detect 1018/110 (overlapping) -  
State diagram -



Moore fsm to detect 101 & 110 (overlapping)

State diagram -

